

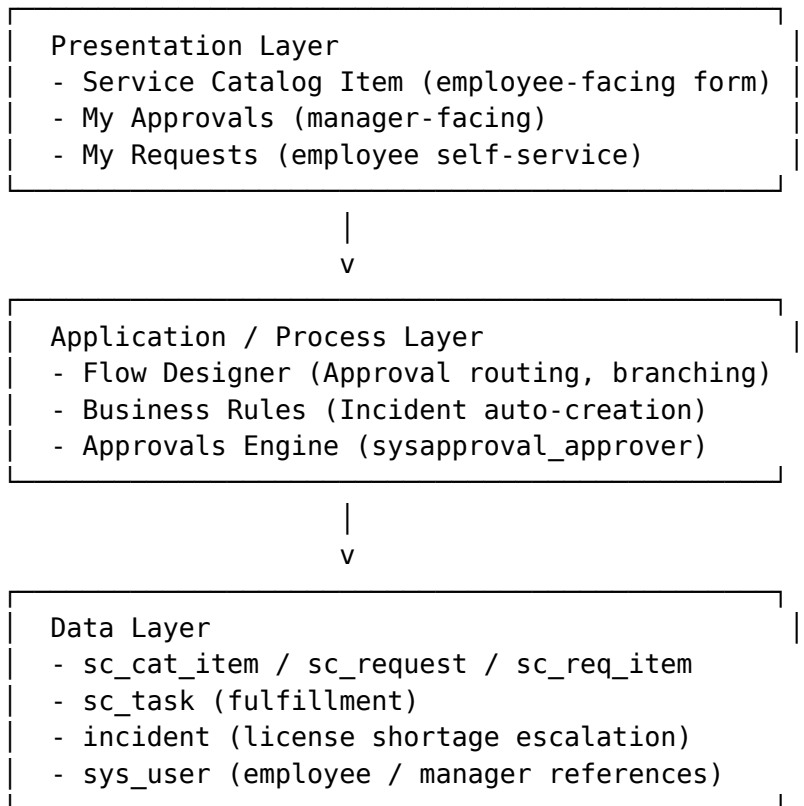
Solution Architecture

Project: Employee Software Installation Request (ServiceNow)

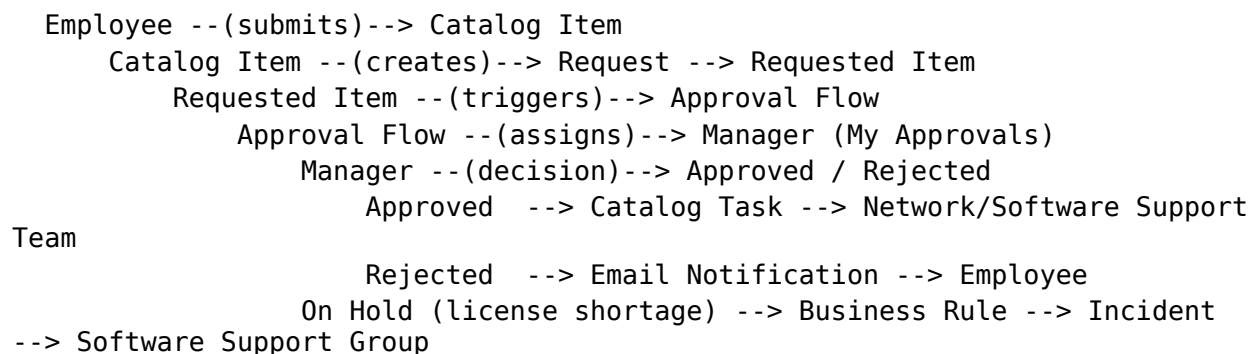
1. Architecture Overview

This solution is built entirely on the ServiceNow platform using a scoped application, following a data-driven workflow architecture: a Service Catalog front-end feeds structured requests into standard Requested Item records, which drive an automated Approval and Fulfillment process via Flow Designer and Business Rules.

2. Layered Architecture



3. Component Interaction Diagram



4. Key Design Decisions

- Built as a scoped application to isolate all custom logic and enable clean packaging/version control.
- Used native Service Catalog + Approvals engine instead of custom tables, to leverage existing, well-tested ServiceNow infrastructure.
- Flow Designer (low-code) used for orchestration instead of classic Workflow Editor, for maintainability.
- Business Rules used only for simple, deterministic logic (e.g., Incident creation on a state change) — kept outside the main Flow to separate concerns.

5. Security Architecture

- Role-based access: Employee, Manager, and Network/Software Support roles define who can view/act on records.
- Approvals are restricted to the specific manager assigned via the Approval Field, preventing unauthorized approval actions.
- Cross-scope access explicitly granted only where required (e.g., email/notification creation from the scoped app).

6. Scalability & Extensibility

- Additional approval levels (e.g., department head) can be added as further branches in the Flow without redesigning the data model.
- A License Inventory table can be introduced later to replace the simple On Hold/Incident escalation with real-time availability checks.
- The same architecture pattern can be reused for other request types (hardware, VPN access, etc.) by creating new Catalog Items pointing to the same Flow pattern.